# Research Statement

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Air pollution, traffic accidents, and groundwater depletion share one feature. Whoever causes the damage does not pay for it. Environmental and resource economics calls this an externality. Governments respond with traffic rules, infrastructure, subsidies, zoning, and certification programs. Whether these policies work, and what else they change, is an important empirical question. My research evaluates them with causal inference and population-wide administrative data from Taiwan, and the results often differ from what the policy assumed. Externalities arise not only from factory chimneys but also on farmland and city streets, so my work follows two connected lines. The first asks whether environmental practices on farms can also raise farm income. The second asks how traffic rules, infrastructure, fares, and large events change accidents and emissions.

## 1. Agricultural Production and the Environment

Practices that lower farm income are unlikely to spread, so this line of work asks whether environmental goals and farm income can go together.

Solar power needs land, but panels on farmland raise concerns about competition with agriculture. Rooftop panels use no farmland, yet evidence on them comes mostly from laboratory studies. Using the 2020 agriculture census, we find that livestock and poultry farms with rooftop panels have higher sales [1], and chicken farms sell 5.8% more, mainly through higher output [2]. Because rooftop space is limited, the government also zones fish ponds for aquavoltaics. Farmland prices rose 13.5% after land entered these zones, which likely reflects the value of the development rights [3].

Beyond energy, the more widespread environmental effect of farming comes from chemical fertilizer and pesticide, which are hard to regulate because they come from many dispersed sources. Food traceability systems target food safety and have been studied mainly from the consumer side. We find that adopting farms are 8.2, 15.4, and 40 percentage points less likely to use chemical fertilizer, pesticide, and groundwater, and their revenue also rises [4]. In other words, a food safety policy also brings environmental benefits.

Not every environmental practice raises income. Forest farms with greater tree diversity earn less on average, but their revenue is much more stable, and the reduction in variability exceeds the loss in income [5]. Environmental practices can thus benefit farmers through lower risk even without higher income.

## 2. Transportation, Mobility, and Externalities

Traffic produces two main externalities, accidents and emissions. Public transit is often seen as a way to reduce both, so I first examine what changes transit use. Taiwan never imposed a COVID-19 lockdown, yet each new confirmed case reduced Taipei Metro ridership by 1.43%, so perceived risk alone changed travel behavior [6]. On the policy side, the Taipei monthly pass raised metro ridership by 4.3% [7]. Emissions fall only if 70% to 80% of new trips replace car or motorcycle trips.

Most trips still rely on motor vehicles, so road safety depends on rules and infrastructure. After part of Tainan City removed the two-stage left turn rule for motorcycles, accidents fell 21% [8]. Counterintuitive rules can raise accident risk. Previous studies of pedestrian protection relied on virtual-reality simulations. We evaluate Taiwan's nationwide intersection upgrade with inverse probability weighting, because the program targeted high-risk intersections. Injury crashes fell 8.2% and pedestrian crashes 14.8%, and each NT$1 spent saved about NT$3.5 in medical costs [9].

Beyond daily travel, large events bring sudden surges of traffic. Air quality fell 17.5% during the Hakka Tomb-Sweeping Day. Previous studies attributed holiday pollution to incense, joss paper, and fireworks, but we find that the main cause is greater motor vehicle use [10]. During the Dajia Mazu Pilgrimage, accidents rose 14.9%, and 9.8% even in neighboring townships off the route [11]. In both daily and religious travel, the externalities come mainly from motor vehicles.

## 3. Consumption and Markets

I also examine how consumers and markets respond to external shocks and policy changes. During the pandemic, consumers updated their beliefs as a Bayesian model predicts and moved food purchases online [12]. After the pandemic, hotel revenue recovered through higher room rates rather than more guests [13]. Markets also adjust when trading stops. Taiwan's largest hog auction market closes every Sunday and on extra days. Festival closures lower traded quantities by about 12% before and after the closure and sharply raise price volatility on the last trading day before it, while closures for supply adjustment have little effect [14]. A strategic trade model also shows that the EU Carbon Border Adjustment Mechanism promotes green investment by exporters [15].

On farms and in cities, my research asks who responds, by how much, and who bears the costs and benefits when a policy or shock changes the cost of harming the environment or the reward for improving it.

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